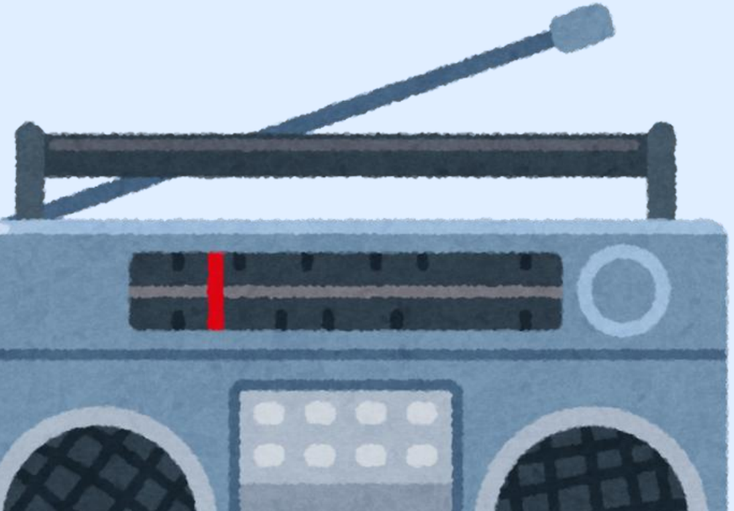




Digital Walkie Talkie

Digital Communication





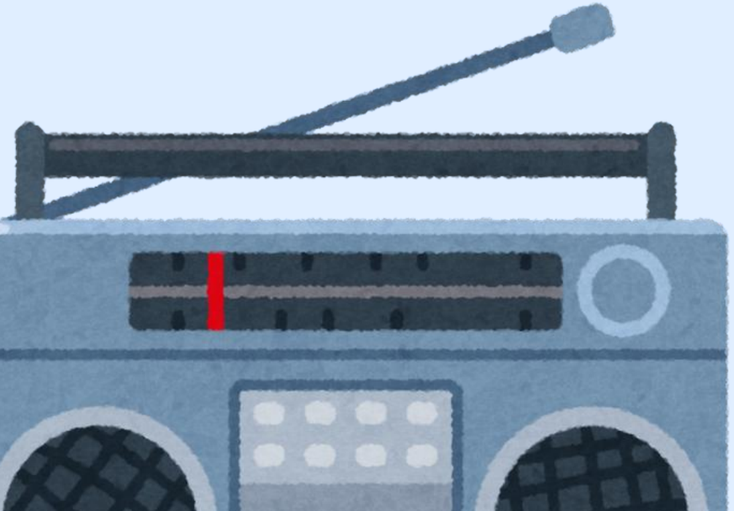






Our Team

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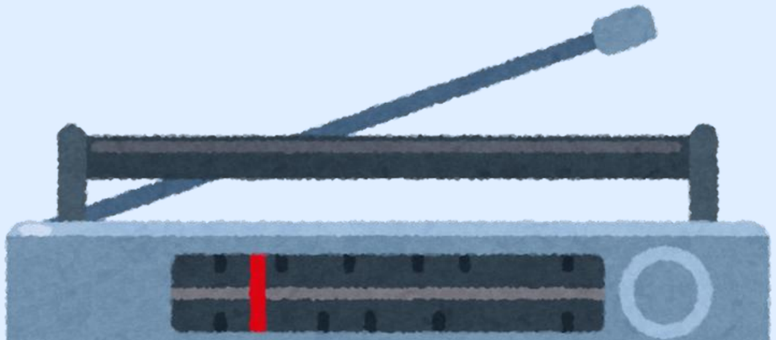






Problem Statement

- Conventional communication methods such as mobile phones depend on cellular networks or internet connectivity. There is a need for a low-cost, portable, short-range communication device capable of transmitting voice directly between two users without depending on cellular infrastructure





Introduction

A digital communication-based walkie-talkie is a wireless device used for sending and receiving voice messages. It converts voice into digital signals and transmits them wirelessly to another device. It allows fast and reliable communication without using a mobile network or internet.

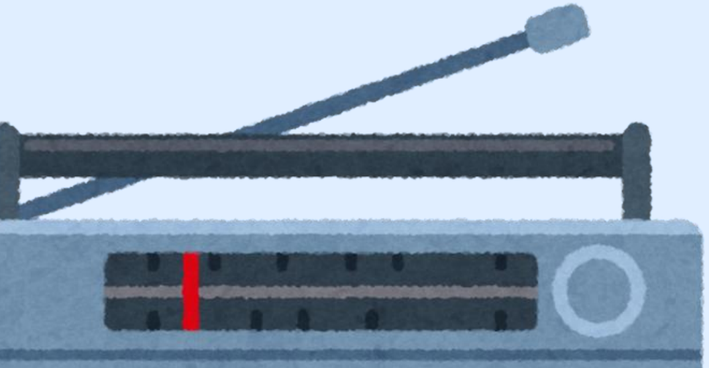


Components Used

- ESP32-S3 – Digital processing and audio handling
- nRF24L01+ – 2.4 GHz wireless transceiver
- INMP441 Microphone – Voice input
- MAX98357 Amplifier– I2S digital audio amplifier
- Speaker – Audio output
- Battery – Portable power supply
- Antenna – RF transmission/reception
- Push Buttons – Push-to-talk and controls
- PCB & wires – Hardware integration









Project Overview

This project is a two-way digital voice communication system using two ESP32 boards and nRF24L01+ transceiver modules. The microphone captures voice, which is converted into digital audio by the ESP32 and transmitted wirelessly using the nRF24L01+. At the receiver, the ESP32 processes the received audio and sends it to the MAX98357 I2S amplifier, which drives the speaker.
Communication technique: GFSK (Gaussian Frequency Shift Keying) — the nRF24L01+ uses GFSK for its RF transmission.

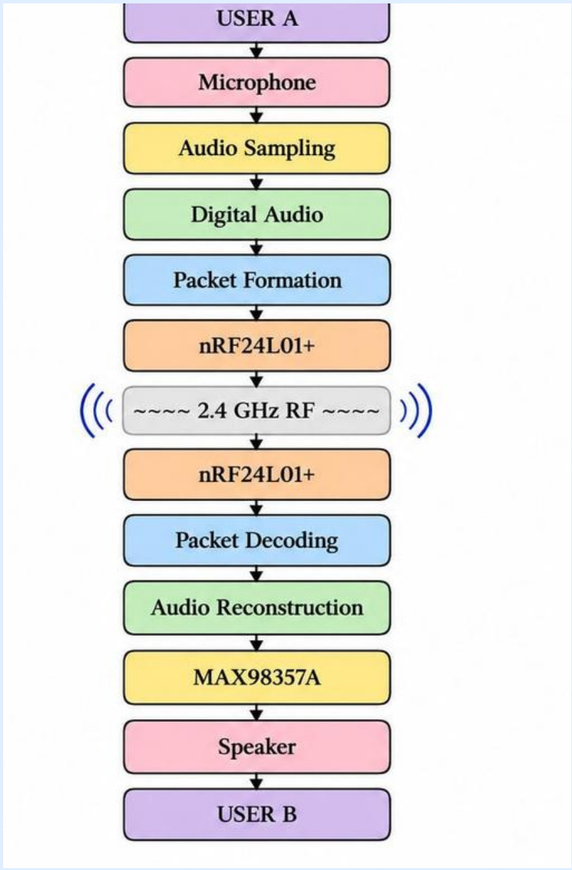




Theory

A digital walkie-talkie is a wireless device used for voice communication without internet or mobile networks. It converts voice into digital signals (0s and 1s) and transmits them using radio waves. The receiver converts the digital signal back into voice through a speaker.

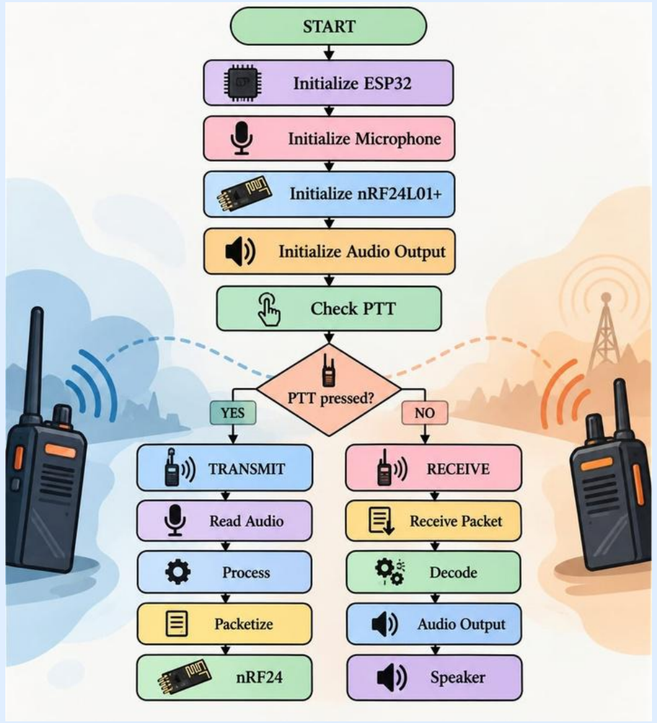
Advantages: Better voice clarity, less noise, reliable communication, and efficient transmission.





Working

Software operation



Data Packet Structure





Applications

- Industrial communication
- Warehouse communication
- Event coordination
- Security teams
- Campus communication
- Disaster/emergency communication
- Remote Area communication

Limitations

- Limited communication range
- 2.4 GHz can be affected by obstacles/interference
- Voice quality depends on audio processing
- Battery capacity limits operating time
- Prototype requires optimization for reliable continuous voice
- Half-duplex communication means only one user talks at a time



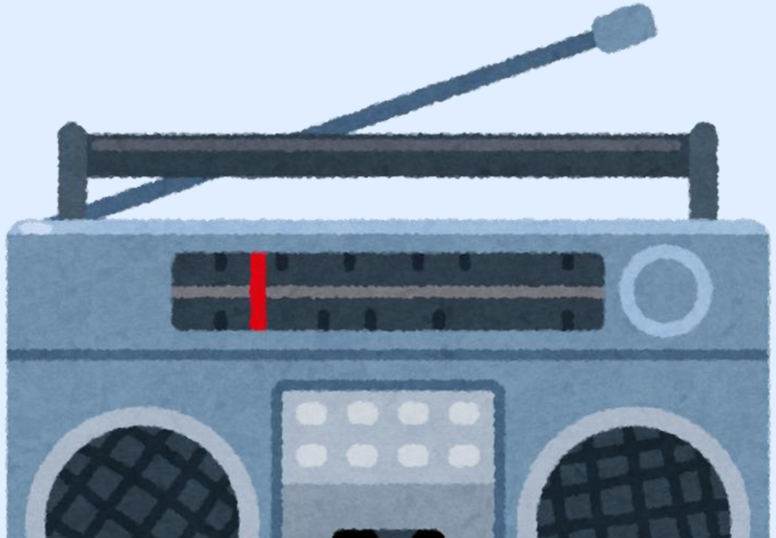






Conclusion

Our project aims to develop a digital communication-based walkie-talkie for simple and reliable wireless voice communication. It converts voice into digital signals and sends them wirelessly from one device to another. The system is expected to provide clear communication without using mobile networks or the internet. It can be useful in industries, security, events, and emergency situations. Since the prototype is still under development, the expected results will be verified through practical testing.









Thank You

